Session 1 — What is AI, and where does it come from?

Artificial Intelligence · 4th year · UIR · Pr. Hakim Hafidi

Study sheet — exam material

1. The thread of this session

This session answers two questions that sound simple and are not: what do we mean by intelligence, and how did people come to believe a machine could have any of it?

The answer runs through twenty-three centuries, from Aristotle's syllogism to the expert systems of the 1980s, and it follows one specific idea: **that reasoning is the manipulation of symbols according to rules, and that a machine can manipulate symbols.** This is the *symbolic* path, also called GOFAI — Good Old-Fashioned AI. It produced the first programs that anyone called intelligent, it ran into two funding collapses known as the AI winters, and it supplied the vocabulary we still use today.

The other path — growing the rules from data instead of writing them — is Session 2. You need both to understand why modern AI looks the way it does.

2. What is intelligence, and why no single definition works

Before asking whether a machine can be intelligent, notice that we cannot define intelligence for ourselves. What we can do is list its facets, and each of them has been attacked separately by machines.

The facets usually listed. Problem solving, planning, learning from experience, using language, creativity, abstract reasoning, perception, social understanding. No single test captures all of them, and a person can be strong in one and weak in another.

Machines took them one at a time. In 1956, a machine could do arithmetic and nothing on the list. By 2026 it can plan, use language, perceive, and — the turning point — learn from experience. Creativity and reasoning remain contested: a system that generates a new image has produced something new, but whether that is creating is a question about meaning, not about output. Reasoning in current systems is fragile in ways Session 5 will make precise.

Why this matters for the course. Every time someone says "AI can now do X", the useful question is: which facet, measured how, under what conditions? That reflex is the whole point of the labs.

3. The ancient idea: reasoning as mechanical procedure

The belief that thought could be mechanised is far older than computers.

Aristotle and the syllogism (~350 BC). Aristotle's contribution was not any particular argument but the discovery of *form*. "All men are mortal; Socrates is a man; therefore Socrates is mortal" is valid because of its shape, not because of what the words mean. Replace men, Socrates and mortal with anything else and the argument still holds. If validity is a property of form, then checking an argument is a mechanical operation — and anything mechanical can in principle be done by a machine.

Al-Khwarizmi and the algorithm (~820). Working in Baghdad, Muhammad ibn Musa al-Khwarizmi wrote procedures for solving equations: finite sequences of unambiguous steps that anyone could follow without understanding why they work. The Latinisation of his name gave us the word *algorithm*, and the title of his book on *al-jabr* gave us *algebra*. The key property is that the procedure's correctness does not depend on the intelligence of whoever executes it — which is exactly what makes it executable by a machine.

Leibniz and the calculus of thought (1670s). Leibniz invented binary arithmetic and built a calculating machine, but his ambition was larger: a universal symbolic language in which every concept could be written, so that disagreements could be settled by computation. His phrase — *calculemus*, "let us calculate" — is the earliest clear statement of the symbolic AI programme.

Pascal and Pascaline (1642). A mechanical adding machine built by a nineteen-year-old Blaise Pascal to help his father with tax calculations. It proved that a physical device could perform an operation previously requiring a mind.

Babbage and Lovelace (1837–1843). Charles Babbage designed the Analytical Engine, a programmable mechanical computer with a store (memory), a mill (processor), and punched-card input. It was never built in his lifetime. Ada Lovelace wrote notes on it that included what is generally considered the first computer program — an algorithm for Bernoulli numbers — and also the first serious statement of scepticism about machine intelligence. "The Analytical Engine has no pretensions whatever to originate anything," she wrote. "It can do whatever we know how to order it to perform." This became known as **Lady Lovelace's objection**, and Turing answered it a century later.

Turing and the universal machine (1936). In *On Computable Numbers*, Alan Turing described an abstract machine with an infinite tape, a read-write head, and a table of rules. He proved that a single such machine, given the right description on its tape, can simulate any other. This is the idea of **software**: hardware need not be rebuilt for each task; the task is data. Everything that follows depends on it.

ENIAC and von Neumann (1945). The first general-purpose electronic computer, and the stored-program architecture that still describes the machine in front of you. Turing's abstraction became a physical object, and the question "can machines think?" stopped being purely philosophical.

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4. The Turing test: defining intelligence by behaviour

In 1950, in *Computing Machinery and Intelligence*, Turing confronted the question directly — and refused it. "Can machines think?" he argued, is too vague to answer, because we cannot define thinking. He replaced it with something testable.

The imitation game. An interrogator exchanges written messages with two hidden participants, one human and one machine. If the interrogator cannot reliably tell which is which, the machine passes. Nothing is said about the machine's internal mechanism: the test is deliberately, explicitly **behavioural**.

What Turing was doing. He was making a methodological move, not a metaphysical claim. If we cannot inspect thinking directly, we can at least agree on an observable criterion. This is the same move that underlies every benchmark you will meet in this course — and it carries the same weakness.

Turing's answers to the objections. The paper anticipates nine objections and answers each. Against Lady Lovelace's objection he points out that machines regularly surprise their programmers, and that a learning machine would produce behaviour nobody wrote. Against the argument from consciousness, he notes that we grant other people minds on behavioural evidence alone.

Criticisms that matter today. The test rewards imitation, including imitation of human error and human ignorance — a machine that answered arithmetic too fast and too correctly would fail. It measures deception rather than capability. And modern language models produce convincingly human text while failing at tasks a child handles, which suggests the test measures something narrower than Turing hoped. Session 3 returns to this with Searle's Chinese Room.

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5. Dartmouth 1956: the field gets a name and a bet

In the summer of 1956, John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon organised a two-month workshop at Dartmouth College. The proposal contains the founding sentence of the field:

> *"Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."*

Two things to notice. First, the term **artificial intelligence** was coined for this proposal, partly to distinguish the new field from cybernetics. Second, the sentence is a *bet*, not a result — and the whole history that follows is the testing of that bet.

The symbolic programme. If intelligence can be precisely described, then intelligence is the manipulation of descriptions. In practice this means three components:

Component	What it is	Who writes it
Knowledge base	Facts about the world, as symbols	A human
Rules	How facts combine to produce new facts	A human
Search / inference	Exploring combinations to reach a goal	The program

The human supplies the first two; the machine does the third. This division is the signature of symbolic AI, and its eventual failure point.

Early successes. The *Logic Theorist* (Newell and Simon, 1956) proved theorems from *Principia Mathematica*, one of them more elegantly than the original. The *General Problem Solver* (1957) separated the problem-solving strategy from the problem description — an ambitious attempt at a universal method. *SHRDLU* (Winograd, 1970) held a genuine conversation about a world of coloured blocks, resolving pronouns and answering questions about its own actions. *Shakey* (SRI, 1966–72) was the first mobile robot to plan a sequence of actions and reason about the consequences.

ELIZA and the effect named after it (1966). Joseph Weizenbaum wrote a 200-line program that imitated a Rogerian psychotherapist by pattern-matching on the user's sentences and turning them into questions. It understood nothing. Users nonetheless confided in it, and Weizenbaum's own secretary asked him to leave the room. The **ELIZA effect** — reading understanding and intention into a system that merely

produces plausible text — is the single most useful concept from this session for thinking about modern chatbots.

6. Why the symbolic path stalled: two winters

Progress was real, then it stopped. Twice.

Combinatorial explosion. Search works by exploring possibilities, and the number of possibilities grows faster than any machine can follow. Chess has roughly 10^{120} possible games — more than there are atoms in the observable universe. Micro-worlds like SHRDLU's blocks were tractable precisely because they were tiny; nothing in the approach scaled to the open world.

No common sense. Symbolic systems knew exactly what they had been told and nothing else. They could not infer that water is wet, that people dislike pain, or that a dropped glass breaks. Attempts to write down common sense explicitly — the Cyc project, begun in 1984 — consumed decades and never reached the goal.

The first winter (1974–1980). In 1973 the British government commissioned the **Lighthill report**, which concluded that AI had failed to deliver on its promises and that its methods would not scale. Funding in the UK was cut almost entirely; American agencies followed. The proximate cause was over-promising: researchers had predicted human-level machine intelligence within a generation.

The expert systems boom (1980–1987). The field returned with a narrower, more commercial proposition. Instead of general intelligence, encode the knowledge of one expert in one domain as IF–THEN rules. **MYCIN** (Stanford, 1976) diagnosed blood infections with about 600 rules and outperformed junior doctors. **XCON** (Digital Equipment Corporation, 1980) configured computer orders using several thousand rules and reportedly saved the company tens of millions of dollars a year. A whole industry grew around specialised Lisp machines.

The second winter (1987–1993). It collapsed for a reason internal to the approach, and this is the one to remember: the **knowledge-acquisition bottleneck**. Experts cannot write down what they know. Much expertise is tacit — a radiologist recognises a pattern without being able to state the rule. Worse, a system of several thousand interacting rules becomes unmaintainable: adding a rule breaks three others, and nobody can predict which. When the Lisp machine market collapsed in 1987, the industry went with it.

The lesson that produced modern AI. If humans cannot write the rules, the rules must come from somewhere else. That somewhere is data — which is Session 2.

7. Key terms

Term	Meaning
Algorithm	A finite list of unambiguous steps that solves a class of problems; named after Al-Khwarizmi (~820).
Syllogism	A form of reasoning whose validity depends on its shape, not on the meaning of its words (Aristotle).
Universal machine	Turing's 1936 abstract machine — tape, head, rules — able to simulate any other machine. The origin of software.
Turing test / imitation game	A behavioural test of intelligence (1950): if an interrogator cannot distinguish machine from human in conversation, the machine passes.
Lady Lovelace's objection	"The Analytical Engine has no pretensions whatever to originate anything" (1843). Turing answers it in 1950.
Symbolic AI / GOFAI	Intelligence as the manipulation of symbols by explicit rules: knowledge base + rules + inference.
Search	Exploring a space of possible states to find a path to a goal (Logic Theorist, General Problem Solver).
ELIZA effect	The tendency to read understanding and feeling into a program that produces plausible text (Weizenbaum, 1966).
Micro-world	A small, completely described environment (SHRDLU's blocks) in which symbolic AI works — and outside which it does not.
Combinatorial explosion	The number of possibilities grows so fast that exhaustive search becomes impossible (chess: ~10^{120} games).
Expert system	A program applying hundreds or thousands of IF–THEN rules elicited from human experts (MYCIN 1976, XCON 1980).
Knowledge engineer	The person who interviews experts and turns their knowledge into rules.
Knowledge-acquisition bottleneck	The difficulty of extracting, encoding and maintaining expert knowledge as rules — the cause of the second winter.
AI winter	A period of collapsed funding and interest following over-promising (1974–1980, 1987–1993).
Lighthill report	The 1973 UK report whose negative assessment triggered the first winter.
Cyc	A project begun in 1984 to encode common sense by hand; illustrates the scale of the problem.

8. Dates and numbers to know

~350 BC Aristotle's syllogism · ~820 Al-Khwarizmi, the algorithm · 1642 Pascaline · 1670s Leibniz, *calculemus* · 1837 Analytical Engine designed · 1843 Lovelace's notes and first program · 1936 Turing machine · 1945 ENIAC and the von Neumann architecture · 1950 Turing test · **1956 Dartmouth workshop, the term "artificial intelligence"** · 1956 Logic Theorist · 1957 General Problem Solver · 1966 ELIZA · 1966–72 Shakey · 1970 SHRDLU · 1973 Lighthill report · **1974–1980 first AI winter** · 1976 MYCIN (~600 rules) · 1980 XCON (several thousand rules) · 1984 Cyc begins · **1987–1993 second AI winter** · chess ~10^{120} possible games.

9. Common misconceptions

"The Turing test measures whether a machine is intelligent." It measures whether a machine can be mistaken for a human in text conversation. Those are different claims, and the gap between them is where most popular confusion about AI lives.

"Symbolic AI failed." It stalled as a route to general intelligence. Its techniques — search, planning, logic, knowledge graphs — are in production everywhere, and Sessions 7 and 8 wire them back around neural models.

"The AI winters were caused by technical failure." They were caused by the distance between promise and delivery. The technology kept improving through both; the funding did not.

"ELIZA fooled people because it was sophisticated." It fooled people because *people* fill in meaning. That asymmetry has not changed.

10. Pause & recall — answers

I. (a) Learning from experience, language, planning — any three facets that moved from *no* to *yes* between 1956 and today. (b) Spam filtering, route planning, recommendation, face unlock, autocomplete. (c) Creativity, because generating is not obviously creating; and reasoning, which remains fragile.

II. (a) It shows that the validity of an argument depends on its form, so reasoning can be a mechanical procedure independent of meaning. (b) A finite sequence of precise, unambiguous steps solving a class of problems. (c) That a machine only does what it is programmed to do — debatable today, since a learning system produces behaviour nobody wrote explicitly.

III. (a) 1936: a mathematical model of computation. 1950: a behavioural test of intelligence. (b) What happens inside — the mechanism. (c) It made general-purpose computation physical, turning a philosophical question into an engineering one.

IV. (a) That every aspect of intelligence can be described precisely enough to be simulated by a machine. (b) Because its replies sounded like understanding; the ELIZA effect is our reading of a mind into text. (c) Knowledge base, rules, and search — a human writes the first two, the program performs the third.

V. (a) Combinatorial explosion, absence of common sense, and the funding cuts that followed the Lighthill report. (b) Rules elicited from experts and applied by an inference engine; thousands of interacting rules become brittle and unmaintainable. (c) Because experts cannot state what they know, so building and updating rule bases does not scale.

11. Sample exam questions

1. Explain the Turing test, then give one argument for and one against its relevance today.
2. Describe the symbolic approach to AI with a concrete example, and explain precisely why it stalled.
3. Why do historians speak of two AI winters? Give the specific cause of each.
4. What is the knowledge-acquisition bottleneck, and in what sense does machine learning answer it?
5. Lady Lovelace wrote that a machine can only do what we order it to perform. State Turing's reply, then say whether a modern language model settles the question.